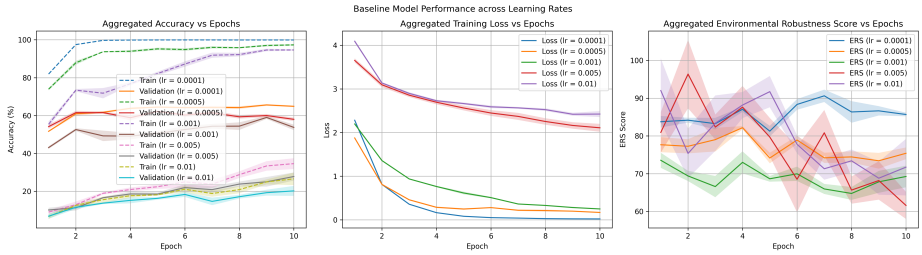

Figure 1: Baseline model performance across learning rates. Aggregated training and validation accuracy, training loss, and Environmental Robustness Score (ERS) over epochs for different learning rates. Lower learning rates yield higher validation accuracy and more stable ERS scores, indicating better generalization and robustness.

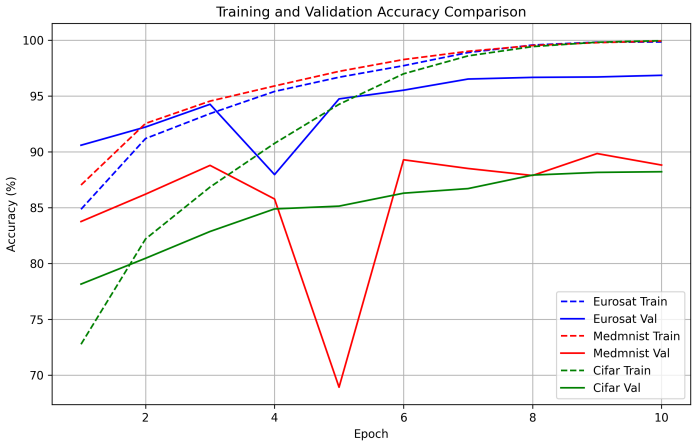

Figure 2: Comparison of training and validation accuracy across different datasets. Models trained on EuroSAT and CIFAR-10 exhibit stable and high accuracy, whereas the model trained on MedMNIST shows erratic accuracy patterns, indicating challenges in generalization due to domain discrepancies.

Comment: Should explain that this is more about generalization from ImageNet to EuroSAT, MedMNIST, or CIFAR.

5.2 Challenges in Multi-Dataset Training

To investigate model generalization across different domains, we trained the ResNet-18 model on additional datasets: EuroSAT, MedMNIST, and CIFAR-10. Figure 2 presents the comparison of training and validation accuracy across these datasets.

From Figure 2, the models trained on EuroSAT and CIFAR-10 achieve high and stable training and validation accuracy over epochs, suggesting effective learning and better generalization. In contrast, the MedMNIST model displays fluctuating accuracy, highlighting difficulties in adapting to the pest detection task. This suggests that significant domain shifts can negatively impact learning, leading to overfitting and decreased robustness.

Comment: a stretched argument. Difficulty of adopting ResNet-18 to MedMNIST doesn't immediately suggest it's also difficult to adapting to the pest detection task. At least it needs more explanation.

To further examine the robustness of these models, we analyzed the ERS across epochs, as shown in Figure 3.

Figure 3 illustrates that the models trained on EuroSAT and CIFAR-10 maintain higher ERS scores across epochs, suggesting robustness to environmental augmentations applied during testing. The MedMNIST model's low ERS scores indicate vulnerability to such changes, underscoring the challenges posed by domain differences. These findings highlight the varying impact of dataset characteristics on model generalization and robustness. While multi-dataset training and domain adapta-

Comment: Should refer to Appendix B, as it provides more detailed information on the noise used